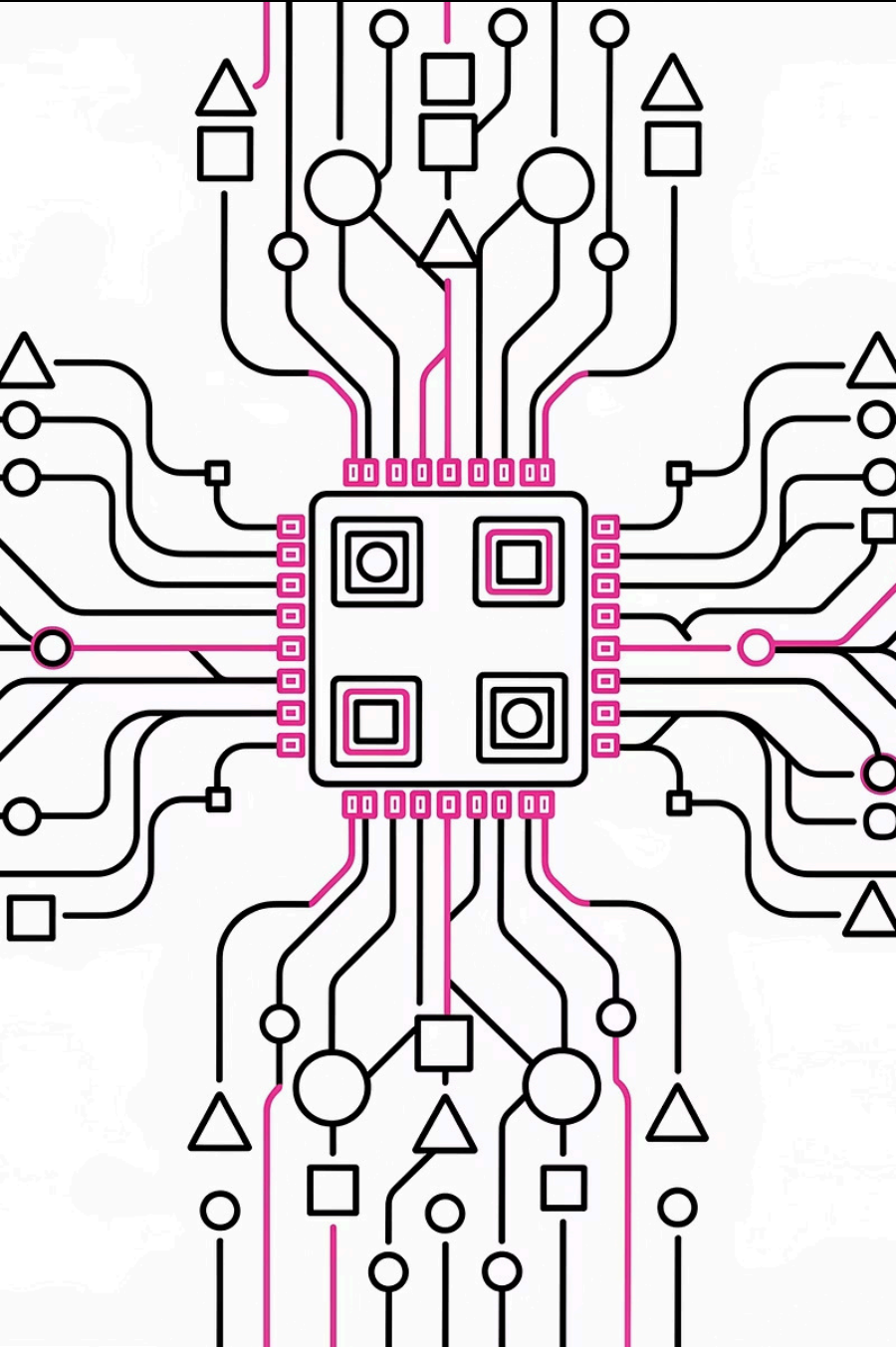




Incorporating Domain Knowledge in Multi-Fidelity Machine Learning Methods

Presenter: Jacob Colton

Course: CS 4490Z – Western
University

Supervisor: Prof. Agustinus Kristiadi,
Dept. of Computer Science

Coordinator: Prof. Nazim Madhavji,
Dept. of Computer Science

Introduction

→ Large Language Models (LLMs) – A Growing Cost

The proliferation of Large Language Models has introduced unprecedented natural language processing, but at a significant computational and ecological cost. (Pope et al., 2022; Strubell et al., 2019).

→ Small Language Models (SLMs) – A Limited Alternative

Small Language Models function as a lower-latency, cost-effective alternative, but struggle with complex reasoning tasks, limiting their autonomous deployment.

→ The Gap & Novelty

Gap: Existing routing approaches lack proactive uncertainty-aware failure prediction prior to generation.

Novelty: This work predicts SLM failure *before* autoregressive generation using a Sparse Variational Gaussian Process (SVGP) on compressed latent embeddings.

Pope, R., et al. (2022). Efficiently scaling transformer inference. arXiv:2211.05102. / Strubell, E., et al. (2019). Energy and policy considerations for deep learning in NLP. ACL.

Table of Contents

01

Background & Related Work

02

Research Methodology

03

Results

04

What's New?

05

Impact on Theory & Practice

06

Limitations of Results

07

Validation

08

Conclusions

09

Future Work & Lessons Learned

10

Extra Material

Background & Related Work

LLM Inference Bottleneck

LLMs are built on autoregressive **self-attention** mechanisms (Vaswani et al., 2017). These mechanisms scale quadratically in both compute time and memory, causing inference to be computationally expensive at scale.

Speculative Decoding

An architecture which reduces latency by running a draft model in parallel with a larger Oracle model. However, it requires a **high parallel memory overhead** (Leviathan et al., 2023) and cannot adjust costs dependent on query difficulty.

Contextual Multi-Armed Bandits (CMAB)

Avoids parallel overhead but suffers an **"exploration tax"** where the system must fail on a query before learning to route it correctly (Sutton & Barto, 2018). This reactive approach can be problematic in high-stakes production environments.

Vaswani, A., et al. (2017). *Attention is all you need*. NeurIPS. | Leviathan, Y., et al. (2023). *Fast inference from Transformers via speculative decoding*. ICML. | Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT Press.

Research Methodology

1

Step 1: Hidden State Extraction

The draft model is intercepted immediately after prompt processing to extract the final continuous **hidden state**, a 960-dimensional embedding vector containing the model's internal representation of the query.

2

Step 2: PCA Compression

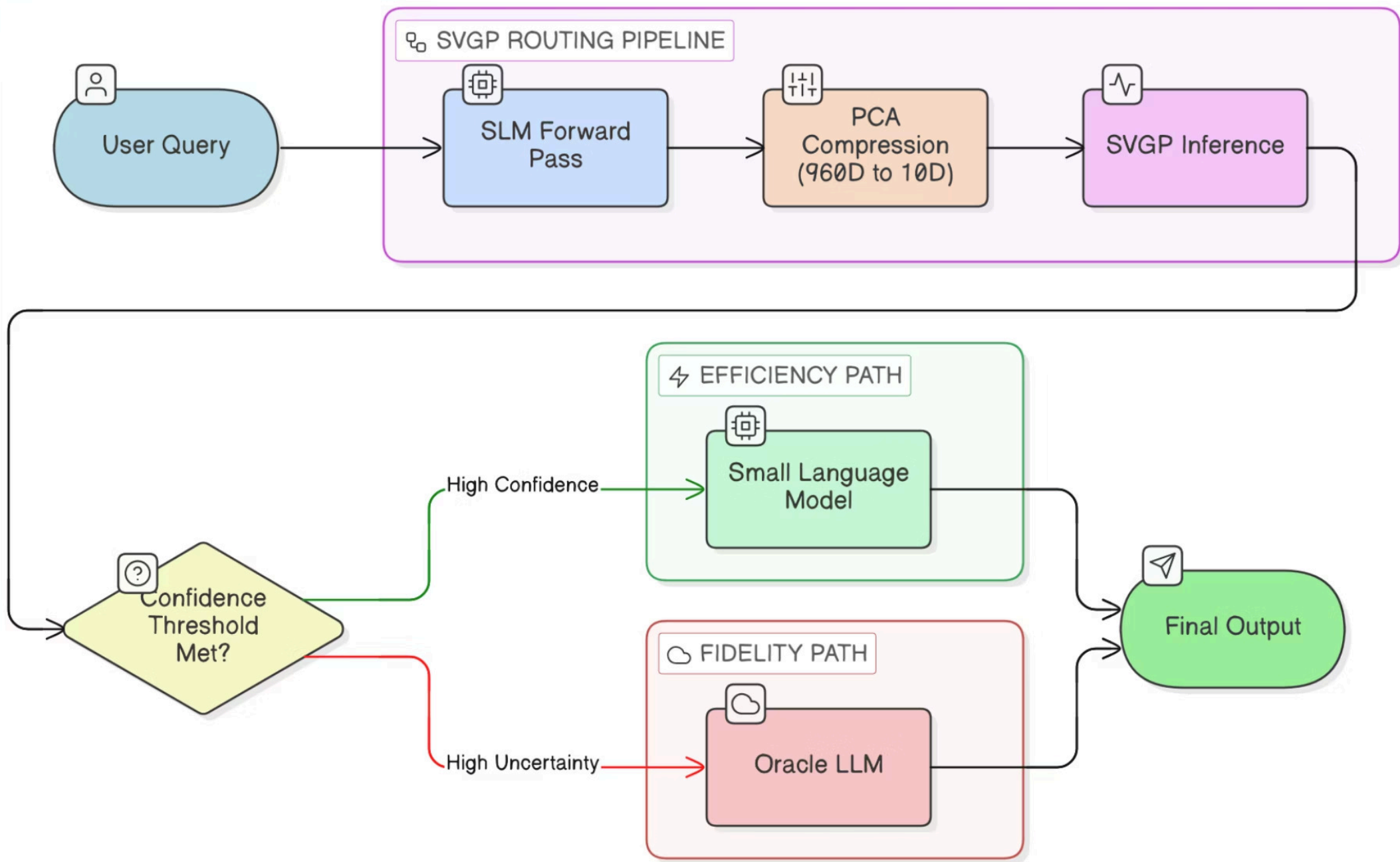
Principal Component Analysis (PCA) compresses the hidden state embedding from 960 dimensions to 10 dimensions while retaining maximal variance ([Pearson, 1901](#)). This reduces dimensionality by preserving the directions of greatest variance.

3

Step 3: SVGP Routing

A **Sparse Variational Gaussian Process** with an ARD-enabled kernel ([Titsias, 2009](#)) allows proactive query routing by classifying each compressed embedding and providing calibrated uncertainty estimates.

Pearson, K. (1901). On lines and planes of closest fit to systems of points in space. Philosophical Magazine. / Titsias, M. (2009). Variational learning of inducing variables in sparse Gaussian processes. AISTATS.



Results – Key Findings



Key Result 1: Proactive Decision Making

Routing decisions are determined **prior to autoregressive generation**. The system successfully predicts likely SLM failure mathematically, allowing for online trial-and-error to be completely bypassed.



Key Result 2: Adjustable Compute

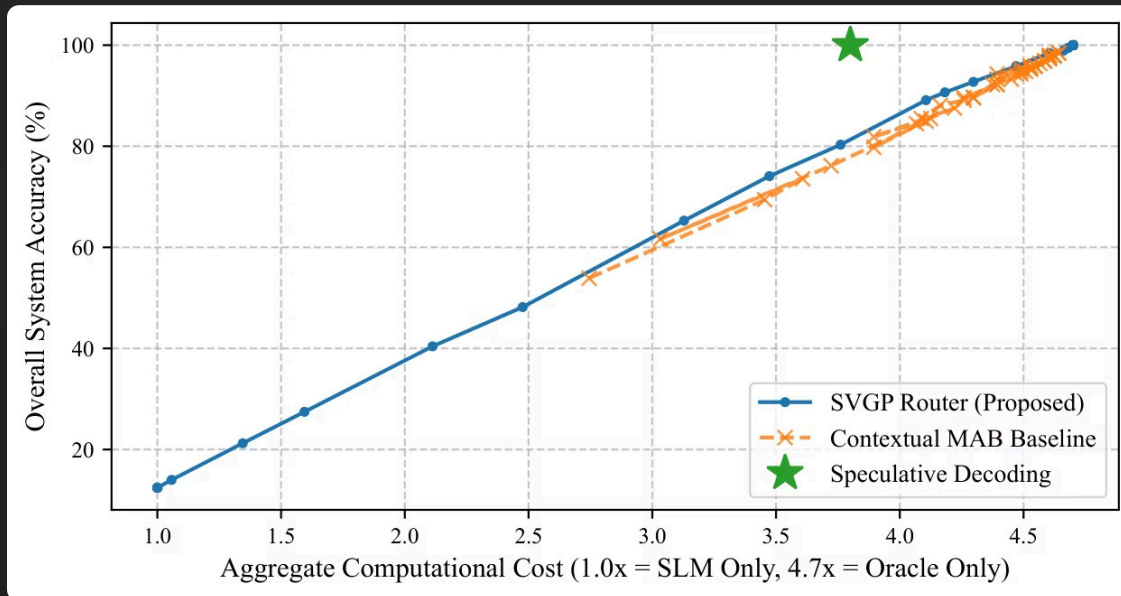
The SVGP provides a **dynamically adjustable computational cost threshold**. This enables system designers to balance accuracy and computational cost in real time without retraining.



Key Result 3: Difficulty Encoding

The SVGP's performance demonstrates that the **SLM's compressed latent space naturally clusters into zones of competence and incompetence**, suggesting that internal representations inherently encode task difficulty.

Results – Code Generation (MBPP)



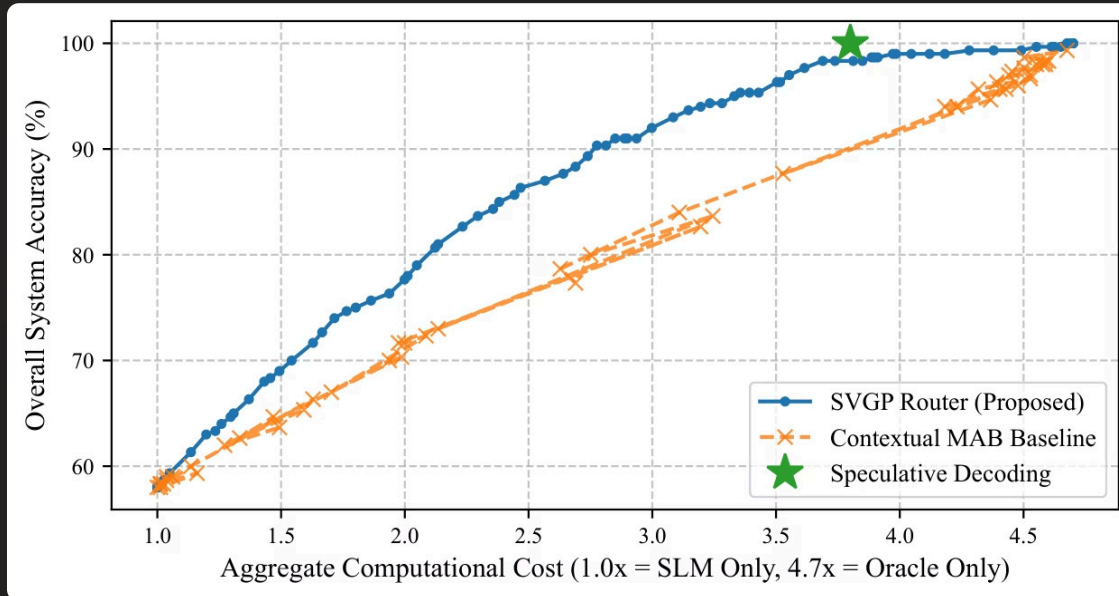
Benchmark: Mostly Basic Python Problems

Evaluated using the MBPP benchmark (Austin et al., 2021), a dataset of entry-level Python programming tasks.

- The SVGP **efficiently maps the entire operational cost range**, from full SLM-only operation to complete Oracle task delegation.
- System designers can set a **precise computational budget** while maximizing overall system accuracy.
- The routing curve remains **consistently above the CMAB baseline**.

Austin, J., et al. (2021). Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*.

Results – Natural Language Processing (MMLU)



Benchmark: Massive Multitask Language Understanding

Evaluated on MMLU (Hendrycks et al., 2020), a dataset spanning **57 distinct academic subjects** from STEM to humanities.

- The SVGP identifies **semantic clusters of competence and incompetence** within the latent space of the SLM.
- Uncertainty estimates are **well-calibrated** across both easy and highly complex topics.

Hendrycks, D., et al. (2020). Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*.

What's New?

Prior Art: Reactive Failure (CMAB)

Systems react **after failure occurs** (e.g., online trial-and-error in CMAB). Each error is a wasted call to the LLM, directly increasing the operational cost and latency of the system.

Prior Art: Brute-Force Parallel Compute

Speculative Decoding requires running both the SLM and the LLM models, incurring a **large fixed overhead** on every query regardless of its actual difficulty.

New: Proactive Latent Space Mapping & Adjustable Cost

This work introduces **proactive mapping** of the SLM's latent space, allowing for mathematical predictions of failure prior to autoregression. No online failures are needed for learning, and the system's computational cost is adjustable.



Impact on Theory & Practice

Theoretical Contribution

- **Geometric Uncertainty:** Demonstrates that AI confidence can be approximated from the geometry of the latent space, entirely prior to text generation.
- **A Mathematical Framework:** Formalizes dynamic routing as a Gaussian Process Regression task, offering a new baseline for multi-model systems.

Practical Impact

Has the potential to reduce the **carbon and financial footprint** of generative AI deployments while maintaining high model accuracy. Helping enable the safe, autonomous deployment of SLMs on **consumer hardware**.

- **Lower energy consumption** per inference
- **Supports more compute-efficient deployment** than other parallel verification frameworks

Limitations of Results

Benchmark Scope – MBPP

MBPP is a **highly controlled sandbox** consisting of simplistic, short Python problems. It is not representative of large software repositories with many files, or real-world engineering tasks with ambiguous specifications.

Benchmark Format – MMLU

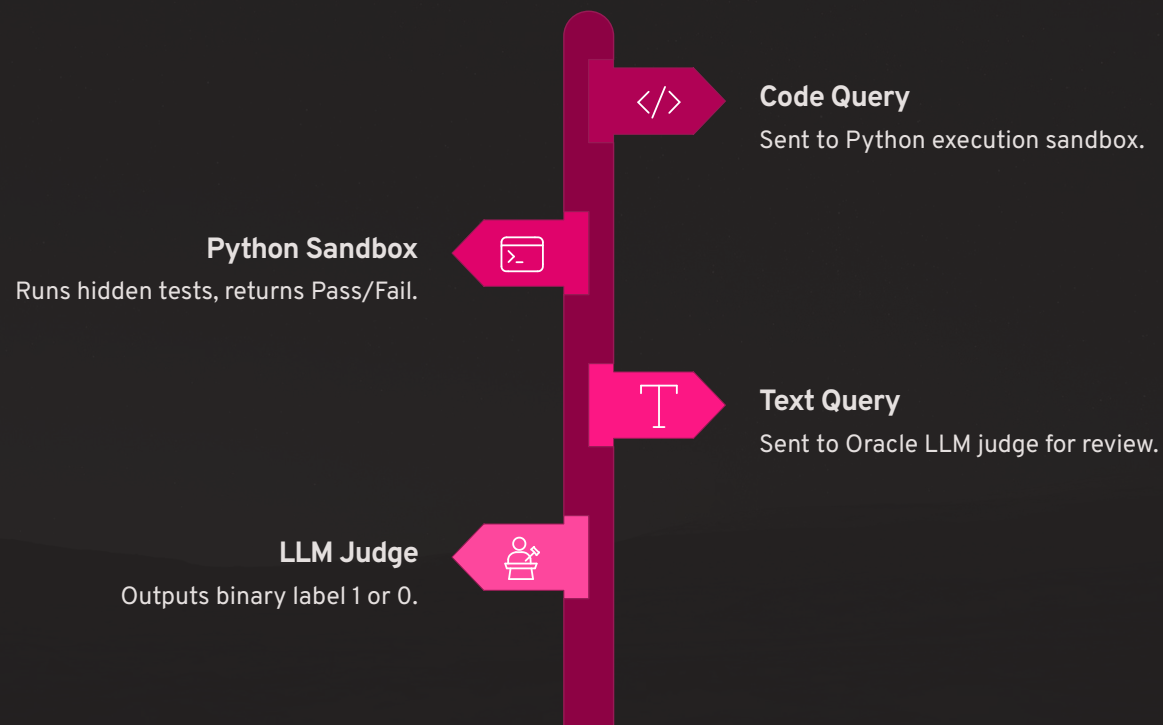
MMLU utilizes a **rigid multiple-choice answer format**, which may not reflect open-ended generative performance. General NLP queries may be more difficult for the SVGP to accurately model.

Dimensionality Reduction – PCA

PCA compression utilizes a **linear method** to shrink data, which causes it to miss intricate, non-linear connections. Due to this, its effectiveness relies entirely on the SLM's task-relevant structure being preserved under a linear projection.



Validation



Code Validation – MBPP

To eliminate any subjective human bias, the model-generated solutions for the MBPP dataset were run within an isolated Python execution sandbox. The Oracle utilized a strict binary pass/fail metric based purely on execution success.

NLP Validation – MMLU

MMLU responses were graded utilizing pointwise binary classification. An Oracle LLM judge was used to convert the free-form SLM response into a binary correct/incorrect label.

Conclusions

Fulfillment of Core Research Objectives

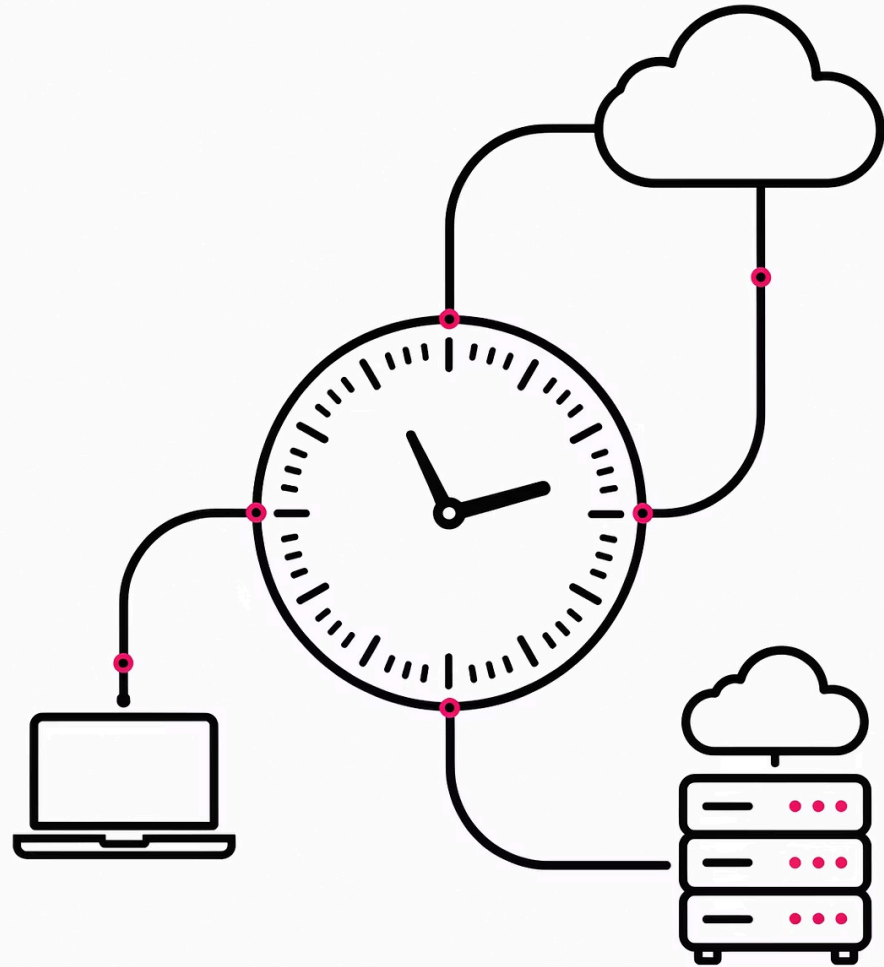
The system successfully formalized multi-fidelity routing as a **Gaussian Process Regression task**, greatly reduced subjective scoring bias utilizing a ground-truth-constrained Oracle framework, and mapped latent competency boundaries.

Calibrated Uncertainty via SVGP

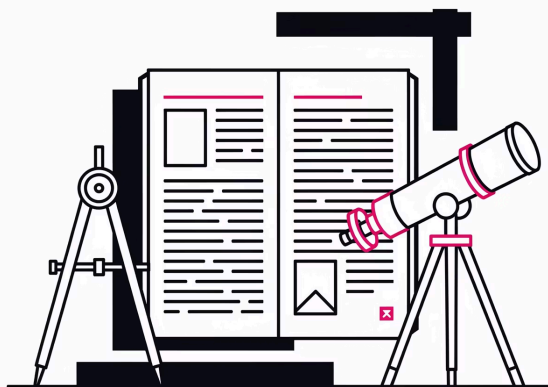
The empirical results confirm the SVGP accurately identifies when the draft model is likely to fail, offering a **dynamically adjustable alternative** option to the fixed computational costs of standard verification frameworks.

Scalable, Sustainable AI

This research provides a **scalable and generalizable** system which can reduce the computational and ecological footprint of generative AI without sacrificing reliability and task accuracy.



Future Work & Lessons Learned



Future Work

- Evaluate utilizing **non-linear dimensionality reduction** (e.g., **UMAP**) to better preserve the non-linear relationships within the language embeddings.
- Test on **open-ended NLP queries**, as opposed to strict multiple-choice formats.
- Explore the implementation of **online SVGP updating** to adapt routing to distribution shift over time.



Key Lessons

- **Lesson 1: 10 Dimensional PCA compression preserves sufficient semantic meaning** for decision-making, challenging the requirement for massive embedding spaces.
- **Lesson 2: Proactive routing is mathematically viable** using only the internal representations of the localized draft model, allowing the Oracle LLM to be treated as a black box.

Architecture & Execution Trace

Query Input

Latent Extraction

PCA 960D→10D

SVGP Uncertainty

References

Austin, J., et al. (2021). Program synthesis with large language models. *arXiv:2108.07732*.

Berger, J. O. (1985). *Statistical decision theory and Bayesian analysis*. Springer.

Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.

Chen, L., et al. (2023). FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv:2305.05176*.

Cobbe, K., et al. (2021). Training verifiers to solve math word problems. *arXiv:2110.14168*.

Do, B., & Zhang, R. (2023). Multi-fidelity Bayesian optimization: A Review. *arXiv:2311.13050*.

Hendrycks, D., et al. (2020). Measuring massive multitask language understanding. *arXiv:2009.03300*.

Kwon, W., et al. (2023). Efficient memory management for large language model serving with PagedAttention. *SOSP*.

Leviathan, Y., et al. (2023). Fast inference from Transformers via speculative decoding. *ICML*.

Pearson, K. (1901). On lines and planes of closest fit to systems of points in space. *Phil. Mag.*, 2(11), 559–572.

Pope, R., et al. (2022). Efficiently scaling transformer inference. *arXiv:2211.05102*.

Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian processes for machine learning*. MIT Press.

Strubell, E., et al. (2019). Energy and policy considerations for deep learning in NLP. *ACL*.

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT Press.

Titsias, M. (2009). Variational learning of inducing variables in sparse Gaussian processes. *AISTATS*.

Vaswani, A., et al. (2017). Attention is all you need. *NeurIPS*.

Wang, Z., et al. (2021). Multi-fidelity high-order Gaussian processes for physical simulation. *AISTATS*.